

AIML Foundations & Python Setup

Day 1: AI vs ML vs DL — An Interactive Guide

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1 What is Artificial Intelligence (AI)?

Definition

Artificial Intelligence (AI) means making computers perform tasks that normally require *human intelligence*.

Examples of human intelligence: recognizing faces, understanding language, playing chess, driving a car, and making decisions. If a machine can do these kinds of tasks, it is considered AI.

Example: A rule-based decision

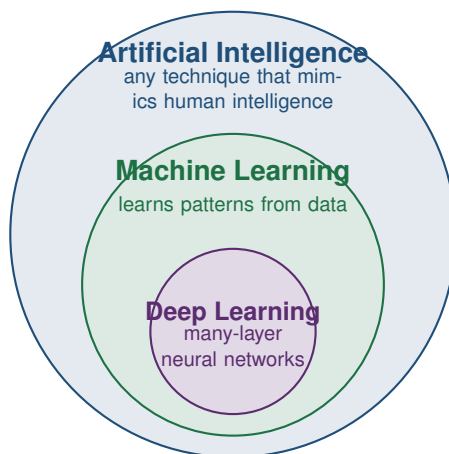
Suppose you create a program that always follows these rules:

```
IF marks >= 40
THEN Pass
ELSE
Fail
```

The computer is making a decision based on rules. This is a simple form of AI.

Real-world AI examples: Google Assistant, Siri, ChatGPT, self-driving cars, chess-playing programs.

1.1 The big picture: $AI \supset ML \supset DL$



2 What is Machine Learning (ML)?

Traditional AI often uses rules written by humans. **Machine Learning is different.** Instead of writing all the rules manually, we give the computer **data**, and it learns patterns from that data.

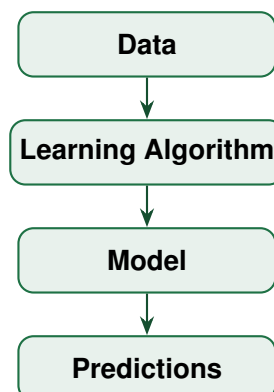
Analogy: Teaching a child

Imagine you show a child many labelled pictures:

Animal	Label
Dog photo	Dog
Dog photo	Dog
Dog photo	Dog
Cat photo	Cat
Cat photo	Cat

After seeing many examples, the child learns that dogs have certain features and cats have certain features. Now if you show a *new* picture, the child can guess whether it is a dog or a cat. **ML works in exactly the same way.** Nobody sat you down and said “a dog has four legs AND fur AND a tail, therefore it’s a dog” — you were shown dogs again and again until your brain built its own internal rule.

2.1 The ML process



Example: Learning from study hours

Hours Studied	Result
2	Fail
4	Pass
6	Pass
8	Pass

The ML algorithm learns a relationship between study hours and result. Now if someone studies 5 hours: **Model** → **likely Pass**. No rule was explicitly written — the machine *learned* it from data.

Example: Spam detection — rules vs. learning

The traditional way: you'd write rules yourself, like "if the email contains 'FREE MONEY' → spam" or "if sender is unknown AND has attachments → spam". You'd need hundreds of such rules, and spammers would quickly find ways around them.

The ML way: collect 10,000 emails, label each as "spam" or "not spam", hand them to an ML model, and it discovers patterns on its own — things a human would never think to write explicitly.

Common ML applications: spam detection, credit scoring, stock prediction, recommendation systems, fraud detection.

Check Your Understanding

(click "Show answer" to jump)

Q1. ML differs from traditional programming because. . . (complete the sentence in one line, then check).

Show answer ↓

Q2. You study 5 hours, and the model from the study-hours table predicts your result. What does it predict and why?

Show answer ↓

3 The Three Types of Machine Learning

Think of ML like three different ways a student can learn. Each one works differently depending on what kind of help is available.

3.1 Type 1 — Supervised Learning

Analogy: Textbook with an answer key

Imagine a student studying for an exam using a textbook that has all the answers at the back. They read a question, write their answer, check the answer key, and correct their mistake. They do this for 10,000 questions. By the end, they've learned the pattern. That "answer key" is the **label**.

Every training example comes with the correct answer already attached. The model's job is to learn to predict that answer for data it hasn't seen before. Supervised learning is the most commonly used type in real life, and it splits into two sub-types:

- **Classification** — the answer is a *category* (spam/not spam, cat/dog, disease/healthy)
- **Regression** — the answer is a *number* (house price, temperature tomorrow, exam score)

3.2 Type 2 — Unsupervised Learning

Analogy: Articles with no labels

A student is given 10,000 news articles but no labels — nobody has told them what category each article belongs to. They start reading and naturally notice that some articles talk about cricket and Virat Kohli, some talk about stock markets and RBI, some talk about films and actors. They create their own groups. **Nobody told them the groups — they discovered them.**

No labels. The model finds hidden structure in the data on its own.



3.3 Type 3 — Reinforcement Learning

Analogy: Learning to ride a bicycle

Nobody hands a child a textbook of labelled examples for cycling. They just try, fall, try again, fall less, try again, and eventually stop falling. Every fall is a **penalty**. Every smooth ride is a **reward**. Over time, the brain learns exactly what to do to get more rewards.

The model (called an **agent**) is placed in an environment. It takes actions, and the environment gives back a reward or a penalty. There's no dataset — the agent generates its own experience by trying things.

3.4 Quick comparison

	Supervised	Unsupervised	Reinforcement
Data available	Yes, labelled	Yes, no labels	No dataset
What it learns	Input → output mapping	Hidden groups / structure	Best action in a situation
Feedback from	Correct labels	None (self-discovers)	Rewards & penalties
Real example	Spam filter	Customer segmentation	Chess AI, self-driving
Hardest part	Getting labelled data	Interpreting what clusters mean	Designing the reward system

Check Your Understanding

(click "Show answer" to jump)

Q3. Predicting tomorrow's temperature is classification or regression?

Show answer ↓

Q4. A telecom company groups its customers by usage behaviour without any labels. Which type of ML is this?

Show answer ↓

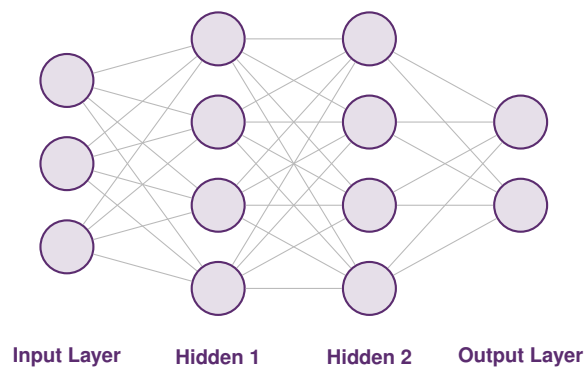
Q5. In reinforcement learning, what replaces the labelled dataset?

Show answer ↓

4 What is Deep Learning (DL)?

Deep Learning is a special type of Machine Learning, **inspired by the human brain**. The brain contains billions of neurons; DL uses artificial neurons organized into layers called **Neural Networks**. When there are many hidden layers, it is called *Deep Learning*.

4.1 Network structure

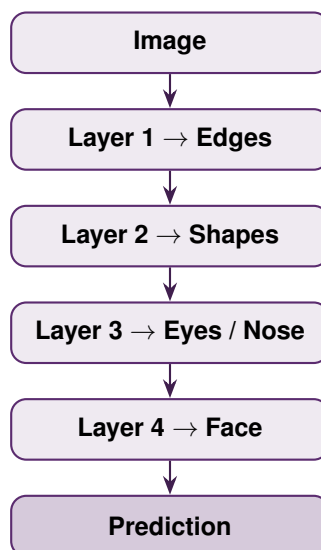


Example: Face recognition — ML vs. DL

Traditional ML: human experts manually define features — eyes, nose, mouth, shape of face.

Deep Learning: give thousands or millions of face images, and the network automatically learns edges → shapes → eyes/nose → complete face. **No manual feature engineering is required.**

4.2 Why “deep”? Layers learn increasingly abstract features



Check Your Understanding

(click “Show answer” to jump)

Q6. What makes a neural network “deep”?

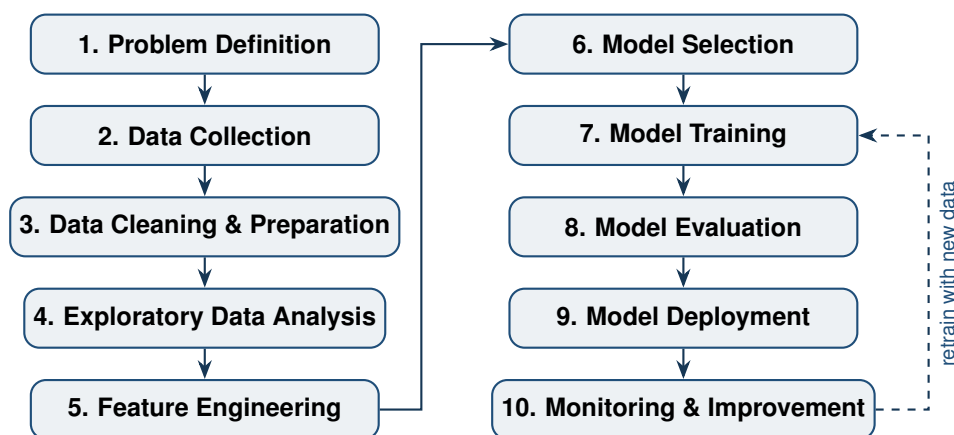
Show answer ↓

Q7. Name the biggest practical advantage of DL over traditional ML for images.

Show answer ↓

5 AI/ML Workflow and Tools Overview

5.1 The 10-step workflow



5.2 Step 1: Problem definition

First, define what problem you want AI to solve.

Problem	AI Task
Predict house prices	Regression
Detect spam emails	Classification
Recognize faces	Computer Vision
Answer questions	Natural Language Processing (NLP)

5.3 Step 2: Data collection

AI learns from data. **Sources:** Excel files, databases, websites, sensors, APIs, images, text documents.

5.4 Step 3: Data cleaning & preparation

Real-world data is messy. Problems include missing values, duplicate records, wrong entries, and outliers. **Tool:** Pandas.

5.5 Step 4: Exploratory Data Analysis (EDA)

Understand the data before training a model. Questions to ask: What patterns exist? Which variables matter? Are there outliers? **Tool:** Matplotlib.

5.6 Step 5: Feature engineering

Convert raw data into useful information.

Example: Splitting a date into features

Original column: Date = 2026-06-12 $\implies$ New features: Year = 2026, Month = 6, Day = 12. These extra features help models learn better. **Tools:** Pandas, NumPy.

5.7 Step 6: Model selection

Choose an algorithm. **Tools:** Scikit-learn, XGBoost.

Problem	Algorithm
House price prediction	Linear Regression
Pass/Fail prediction	Logistic Regression
Classification	Decision Trees
Complex patterns	Random Forest
Image recognition	Neural Networks

5.8 Step 7: Model training

The model learns from historical data (e.g., the hours-studied table from Section 2).

5.9 Step 8: Model evaluation

Check how good the model is by comparing predictions against actual outcomes. **Classification metrics:** Accuracy, Precision, Recall, F1 Score. **Regression metrics:** MAE, MSE, RMSE, R^2 . **Tools:** Scikit-learn, Python.

5.10 Step 9: Model deployment

Make the model available to users.

Example: House price prediction in production

A user enters: Area = 1500 sq ft, Bedrooms = 3. AI predicts: Price = Rs. 50,00,000.

Deployment tools: FastAPI, Flask, Docker, Amazon Web Services, Google Cloud.

5.11 Step 10: Monitoring & improvement

After deployment: collect new data, track errors, retrain models, improve performance. **AI is not “train once and forget” — it requires continuous improvement** (note the dashed feedback arrow in the workflow diagram).

Check Your Understanding

(click “Show answer” to jump)

Q8. Why does the workflow diagram have a dashed arrow looping back from Step 10 to Step 7?

Show answer ↓

Q9. You want to predict whether a transaction is fraudulent. Which step decides whether this is classification or regression, and which is it?

Show answer ↓

6 Python Tools Overview

Tool	What it does / when used	Example
NumPy	Fast mathematical calculations and arrays; numerical computations and matrix operations.	Calculating averages, matrix multiplication, large numerical datasets.
Pandas	Organizes data into tables (rows/columns); data loading, cleaning, filtering, analysis.	Reading an Excel file, removing missing values, selecting rows.
Matplotlib Seaborn	/ Graphs and charts for visual understanding; data exploration and presenting results.	Bar charts, histograms, pie charts, scatter plots, heatmaps.
Scikit-learn	Ready-made ML algorithms; train models without implementing from scratch.	House-price prediction with Linear Regression; spam detector.
TensorFlow Keras	/ Deep Learning models for complex tasks, when traditional ML is not enough.	Image recognition, speech recognition, chatbots, large AI systems.

A Quiz Answers

A1. In traditional programming, humans write the rules; in ML, the computer learns the rules (patterns) from labelled data/examples on its own.

[↑ Back to question](#)

A2. Likely **Pass** — the model learned that students studying 4+ hours tend to pass, and 5 lies in that region even though “5” never appeared in the training data. This is called *generalization*.

[↑ Back to question](#)

A3. Regression — the answer is a continuous number, not a category.

[↑ Back to question](#)

A4. Unsupervised learning (specifically, clustering / customer segmentation) — there are no labels; the model discovers groups on its own.

[↑ Back to question](#)

A5. Rewards and penalties from the environment. The agent generates its own experience by taking actions and observing the feedback.

[↑ Back to question](#)

A6. Having **many hidden layers** between the input and output layers. Each layer learns increasingly abstract features (edges → shapes → parts → whole objects).

[↑ Back to question](#)

A7. Automatic feature learning — no manual feature engineering. The network discovers useful features (edges, shapes, parts) directly from raw pixels.

[↑ Back to question](#)

A8. Because deployed models degrade as the real world changes (*data drift*). New data is collected, errors are tracked, and the model is **retrained** — making the workflow a continuous cycle, not a one-time pipeline.

[↑ Back to question](#)

A9. Step 1 (Problem Definition). It is **classification** — the output is a category (fraud / not fraud).

[↑ Back to question](#)